实验五 变异注释

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[实验目的]

1. 了解变异注释的基本概念与其在基因组学分析中的作用
2. 掌握 ANNOVAR 的基本原理与使用方法
3. 完成从原始 VCF 文件到注释报告的全流程分析
4. 解读注释结果中的重要生物信息学指标

[背景介绍]

变异注释在生物信息学中具有重要意义，其能够帮助研究者理解变异对性状的潜在影响。基因组测序技术的迅猛发展使得我们可以在不同生物中识别出多种类型的基因变异，包括单核苷酸突变（单核苷酸多态性，SNP）、小片段插入和缺失（indels）、以及更大规模的染色体水平上的变异（如染色体移位、缺失和替换等）。通过变异注释，可以明确这些变异对功能的潜在影响，并进行功能预测，从而评估其是否与特定疾病、生物学过程或表型特征相关。

[实验方法]

首先对已有的.vcf文件整理得到输入文件，步骤包括将发生在同一位点的不同变异拆分为多行、统一突变位点位置的计算方式和将.vcf 转换为 ANNOVAR 支持的输入文件格式。之后使用 ANNOVAR 进行两次注释。第一次为基因注释：根据参考基因组将所有突变与特定基因关联起来，确定突变的类型（内含子突变、外显子突变或基因间突变），并根据结果统计每个基因的变异次数（这里需要排除基因间突变），找到变异次数最多的基因。第二次为功能注释：根据参考数据得到每个变异对基因功能的潜在影响。

[实验结果]

1. 选用一种在线工具对文件中前 10 条变异进行注释（计 1 分）

这里选择使用 SIFT 注释变异可能的危害性，输入 vcf 文件中前 10 条变异（如图 1 所示），输出结果如图 2 所示。

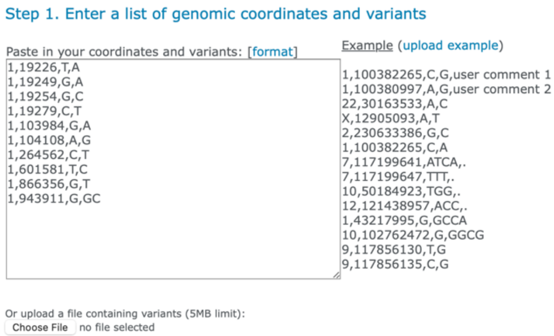

图 1：手动输入突变信息

#ROW_NO.	INPUT	PROTEIN_ID	LENGTH	STRAND	CODON_CHANGE	POS	RESIDUE_REF	RESIDUE_ALT	TYPE	SCORE	PREDICTION (cutoff=-2.5)	#SEQ	#CLUSTER	SCORE	PREDICTION (cutoff=0.05)	MEDIAN_INFO	#SEQ	dbSNP_ID
1	1,19226,T,A	record not found																rs138930629
2	1,19249,G,A	record not found																
3	1,19254,G,C	record not found																rs11525710
4	1,19279,C,T	record not found																rs11525709
5	1,103984,G,A	record not found																
6	1,104108,A,G	record not found																rs4492556
7	1,264562,C,T	record not found																
8	1,601581,T,C	record not found																
9	1,866356,G,T	record not found																
10	1,943911,G,GC	record not found																

#Type	Total	Found_in_dbSNP	Not_found_in_dbSNP	PROVEAN_neutral	PROVEAN_deleterious	PROVEAN_NA	SIFT_tolerated	SIFT_damaging	SIFT_NA
Protein_coding	0	0	0	0	0	0	0	0	0
Single_AA_Change	0	0	0	0	0	0	0	0	0
Synonymous	0	0	0	0	0	0	0	0	0
Deletion	0	0	0	0	0	0	0	0	0
Insertion	0	0	0	0	0	0	0	0	0
Multiple_AA_Change	0	0	0	0	0	0	0	0	0
Frameshift	0	0	0	0	0	0	0	0	0
Nonsense	0	0	0	0	0	0	0	0	0
Unknown	0	0	0	0	0	0	0	0	0
Input_error	0	0	0	0	0	0	0	0	0
Non_protein_coding	10	4	6	0	0	10	0	0	10
Input_format_error	0	0	0	0	0	0	0	0	0

图 2：注释结果

SIFT 结果表明这 10 条变异发生的位置均不编码蛋白，在 SIFT 中无法进行注释。在 dbSNP 数据库中查询以上 10 条变异，发现这些变异大多发生在内含子或两个基因之间。除此之外，SIFT 结果中给出的 dbSNP ID 和.vcf 文件中的不完全一致，检查发现为数据库更新导致。

2. 准备输入文件（计 3 分）

(1) `bcftools norm` 将同一位置的不同变异分行处理

(2) `bcftools norm` 向左标准化

(3) `convert2annovar` 转为 ANNOVAR 接收的 `avinput` 格式

结果如图 3 所示：

```
[teach80_pkuhpc@login66 peiyuliu]# /gpfs1/share/liufenglin/data/class7/software/bcftools/bcftools norm -m-both -o step1.vcf a34e77c6-e8c5-44a6-806f-fd4346284b10.vcf.gz
Lines   total/split/realigned/skipped:  42225/0/0/0
[teach80_pkuhpc@login66 peiyuliu]# /gpfs1/share/liufenglin/data/class7/software/bcftools/bcftools norm -f /gpfs1/share/liufenglin/data/class7/GRCh38.d1.vd1.fa -o step2.vcf step1.vcf
Lines   total/split/realigned/skipped:  42225/0/0/0
[teach80_pkuhpc@login66 peiyuliu]# ./annovar/convert2annovar.pl -format vcf4 -allsample -withfreq ./step2.vcf > ./step3.avinput
perl: warning: Setting locale failed.
perl: warning: Please check that your locale settings:
        LANGUAGE = (unset),
        LC_ALL = (unset),
        LC_CTYPE = "UTF-8",
        LANG = "en_US.UTF-8"
    are supported and installed on your system.
perl: warning: Falling back to a fallback locale ("en_US.UTF-8").
NOTICE: Finished reading 45047 lines from VCF file
NOTICE: A total of 42225 locus in VCF file passed QC threshold, representing 40457 SNPs (19633 transitions and 20824 transversions) and 1768 indels/substitutions
NOTICE: Finished writing allele frequencies based on 80914 SNP genotypes (39266 transitions and 41648 transversions) and 3536 indels/substitutions for 2 samples
```

图 3：准备输入文件

3. 批量注释变异所在基因（计 2 分）

(1) 下载注释文件（图 4）

```
[teach80_pkuhpc@login66 peiyuliu]# ./annovar/annotate_variation.pl -downdb -buildver hg38 -webfrom annovar refGene ./annovar/humandb/
perl: warning: Setting locale failed.
perl: warning: Please check that your locale settings:
        LANGUAGE = (unset),
        LC_ALL = (unset),
        LC_CTYPE = "UTF-8",
        LANG = "en_US.UTF-8"
    are supported and installed on your system.
perl: warning: Falling back to a fallback locale ("en_US.UTF-8").
NOTICE: Web-based checking to see whether ANNOVAR new version is available ... Done
----------------------------UPDATE AVAILABLE------------------------------
--------------------------------------------------------------------------
WARNING: A new version of ANNOVAR (dated 2025-03-02) is available!
         Download from http://www.openbioinformatics.org/annovar/
Changes made in the 2025-03-02 version:
         * turn on -polish by default in table_annovar.pl to fine-tune amino acid changes for coding mutations
         * improve handling of corrupted or incomplete download of ANNOVAR databases during decompression process
         * update the sample files in the genometrax-sample-files-gff directory
         * fixed a bug in URL handling in annotate_variation.pl
         * fixed an issue when calculating protein sequences for mutations affecting multi-mapping transcripts in coding_change.pl
--------------------------------------------------------------------------
--------------------------------------------------------------------------
NOTICE: Downloading annotation database http://www.openbioinformatics.org/annovar/download/hg38_refGene.txt.gz ... OK
NOTICE: Downloading annotation database http://www.openbioinformatics.org/annovar/download/hg38_refGeneMrna.fa.gz ... OK
NOTICE: Downloading annotation database http://www.openbioinformatics.org/annovar/download/hg38_refGeneVersion.txt.gz ... OK
NOTICE: Uncompressing downloaded files
NOTICE: Finished downloading annotation files for hg38 build version, with files saved at the './annovar/humandb' directory
[teach80_pkuhpc@login66 peiyuliu]# ls -lrt ./annovar/humandb/
total 868322
-rw-r--r-- 1 teach80_pkuhpc teach80_pkuhpc  21759357 Jun  8  2020 hg19_refGene.txt
-rw-r--r-- 1 teach80_pkuhpc teach80_pkuhpc    947382 Jun  8  2020 hg19_refGeneVersion.txt
-rw-r--r-- 1 teach80_pkuhpc teach80_pkuhpc 255891394 Jun  8  2020 hg19_refGeneMrna.fa
-rw-r--r-- 1 teach80_pkuhpc teach80_pkuhpc  21791520 Jun  8  2020 hg19_refGeneWithVer.txt
-rw-r--r-- 1 teach80_pkuhpc teach80_pkuhpc 256711379 Jun  8  2020 hg19_refGeneWithVerMrna.fa
-rw-r--r-- 1 teach80_pkuhpc teach80_pkuhpc   2076545 Jun  8  2020 hg19_example_db_gff3.txt
-rw-r--r-- 1 teach80_pkuhpc teach80_pkuhpc      5965 Jun  8  2020 hg19_example_db_generic.txt
-rw-r--r-- 1 teach80_pkuhpc teach80_pkuhpc     23048 Jun  8  2020 hg19_MT_ensGeneMrna.fa
-rw-r--r-- 1 teach80_pkuhpc teach80_pkuhpc      3205 Jun  8  2020 hg19_MT_ensGene.txt
drwxr-xr-x 2 teach80_pkuhpc teach80_pkuhpc      4096 Jun  8  2020 genometrax-sample-files-gff
-rw-r--r-- 1 teach80_pkuhpc teach80_pkuhpc     19843 Jun  8  2020 GRCh37_MT_ensGeneMrna.fa
-rw-r--r-- 1 teach80_pkuhpc teach80_pkuhpc      3131 Jun  8  2020 GRCh37_MT_ensGene.txt
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc  25673940 Aug  3  2022 hg38_refGene.txt
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc   1034484 Aug  3  2022 hg38_refGeneVersion.txt
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 302975562 Aug  3  2022 hg38_refGeneMrna.fa
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc      1966 Mar 25 15:21 annovar_downdb.log
[teach80_pkuhpc@login66 peiyuliu]#
```

图 4:下载参考基因组文件

(2) 注释（图 5），输出文件如图 6 所示

```
[teach80_pkuhpc@login66 peiyuliu]# ./annovar/annotate_variation.pl -out step4 -build hg38 ./step3.avinput ./annovar/humandb/
perl: warning: Setting locale failed.
perl: warning: Please check that your locale settings:
        LANGUAGE = (unset),
        LC_ALL = (unset),
        LC_CTYPE = "UTF-8",
        LANG = "en_US.UTF-8"
    are supported and installed on your system.
perl: warning: Falling back to a fallback locale ("en_US.UTF-8").
NOTICE: The --geneanno operation is set to ON by default
NOTICE: Output files are written to step4.variant_function, step4.exonic_variant_function
NOTICE: Reading gene annotation from annovar/humandb/hg38_refGene.txt ... Done with 88819 transcripts (including 21511 without coding sequence annotation) for 28307 unique genes
NOTICE: Processing next batch with 42225 unique variants in 42225 input lines
NOTICE: Reading FASTA sequences from annovar/humandb/hg38_refGeneMrna.fa ... Done with 16213 sequences
WARNING: A total of 606 sequences will be ignored due to lack of correct ORF annotation
```

图 5:进行注释

```
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc   2836892 Mar 25 15:24 step4.variant_function
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc   1422605 Mar 25 15:24 step4.exonic_variant_function
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc       967 Mar 25 15:24 step4.log
```

图 6:变异注释脚本输出的文件

`step4.variant_function` 的部分内容如图 7 所示，与在 SIFT 和 dbSNP 数据库中查询的结果基本一致，前 10 条变异大多发生在内含子或基因之间。

```
ncRNA_intronic  WASH7P  chr1    19226   19226   T       A       0.25    .       5
ncRNA_intronic  WASH7P  chr1    19249   19249   G       A       0.25    .       6
ncRNA_intronic  WASH7P  chr1    19254   19254   G       C       0.25    .       6
ncRNA_intronic  WASH7P  chr1    19279   19279   C       T       0.25    .       4
intergenic      OR4F5(dist=33976),LOC729737(dist=30789) chr1    103984  103984  G       A       0.25    .       2
intergenic      OR4F5(dist=34100),LOC729737(dist=30665) chr1    104108  104108  A       G       0.25    .       3
intergenic      FAM138D(dist=57965),OR4F29(dist=186178) chr1    264562  264562  C       T       0.25    .       5
intergenic      LOC100132287(dist=106136),LOC101928626(dist=25799)      chr1    601581  601581  T       C       0.25    .       4
intergenic      LINC01128(dist=6910),FAM41C(dist=1715)  chr1    866356  866356  G       T       0.25    .       29
exonic  SAMD11  chr1    943911  943911  -       C       0.25    .       12
```

图 7:变异注释结果

4. 数据可视化与结果解释

(1) 统计基因变异次数

根据 `step4.variant_function` 中的信息进行统计，基因变异次数分布直方图如图 8 所示。

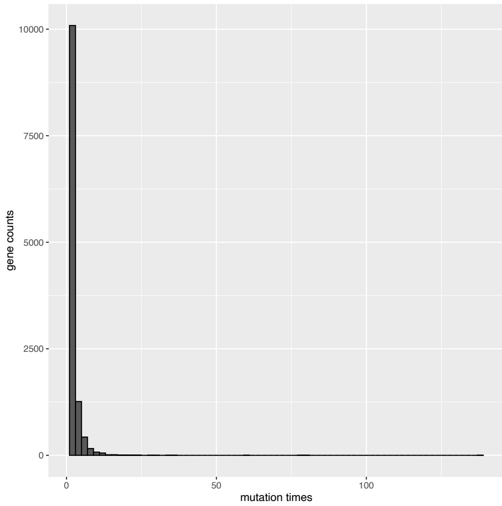

图 8: 基因变异次数分布直方图

(2) 找出突变频数最高的前 10 个基因（图 9）

Var1 <fctr>	Freq <int>
TCAF2	139
BAGE2	81
ROCK1P1	78
MUC6	61
TTN	37
SHANK2	34
NBPF1	31
MUC12	29
LINC01410	25
RYR2	25

1–10 of 12,120 rows   Previous 1 2 3 4 5 6 ... 100 Next

图 9: 突变频数最高的前 10 个基因

(3) 解释某个基因上变异的次数可能与哪些因素有关（计 2 分）

推测某个基因上变异的次数可能与以下几个因素有关：(a)基因的长度：若每个碱基上发生变异的概率相等，则变异次数服从二项分布，即基因的长度越长，期望发生的变异次数越多；(b)基因的表达活性：基因的表达活性越高，DNA 双链解旋的程度越大，提高了变异发生的几率；(c)基因所在的细胞的类型：分裂较旺盛的细胞可能会产生更多的变异。

5. 注释文件中可能有功能的变异（计 2 分）

(1) 将参考数据（hg38_dbnsfp33a）拷贝到 humandb 文件夹，结果如图 10 所示

```
[teach80_pkuhpc@login66 peiyuliu]# ls ./annovar/humandb/
GRCh37_MT_ensGene.txt      genometrax-sample-files-gff  hg19_example_db_generic.txt  hg19_refGeneMrna.fa        hg19_refGeneWithVerMrna.fa  hg38_refGene.txt
GRCh37_MT_ensGeneMrna.fa   hg19_MT_ensGene.txt          hg19_example_db_gff3.txt     hg19_refGeneVersion.txt    hg38_dbnsfp33a.txt          hg38_refGeneMrna.fa
annovar_downdb.log         hg19_MT_ensGeneMrna.fa       hg19_refGene.txt             hg19_refGeneWithVer.txt    hg38_dbnsfp33a.txt.idx      hg38_refGeneVersion.txt
```

图 10:获取参考数据

(2) 进行功能注释（图 11），输出的文件如图 12 所示

```
[teach80_pkuhpc@login66 peiyuliu]# ./annovar/table_annovar.pl ./step3.avinput ./annovar/humandb/ -protocol dbnsfp33a -operation f -build hg38 -nastring .
perl: warning: Setting locale failed.
perl: warning: Please check that your locale settings:
        LANGUAGE = (unset),
        LC_ALL = (unset),
        LC_CTYPE = "UTF-8",
        LANG = "en_US.UTF-8"
    are supported and installed on your system.
perl: warning: Falling back to a fallback locale ("en_US.UTF-8").
NOTICE: the --polish argument is set ON automatically (use --nopolish to change this behavior)
-----------------------------------------------------------------
NOTICE: Processing operation=f protocol=dbnsfp33a
NOTICE: Finished reading 66 column headers for '-dbtype dbnsfp33a'

NOTICE: Running system command <annotate_variation.pl -filter -dbtype dbnsfp33a -buildver hg38 -outfile ./step3.avinput ./step3.avinput ./annovar/humandb/ -otherinfo>
perl: warning: Setting locale failed.
perl: warning: Please check that your locale settings:
        LANGUAGE = (unset),
        LC_ALL = (unset),
        LC_CTYPE = "UTF-8",
        LANG = "en_US.UTF-8"
    are supported and installed on your system.
perl: warning: Falling back to a fallback locale ("en_US.UTF-8").
NOTICE: Output file with variants matching filtering criteria is written to ./step3.avinput.hg38_dbnsfp33a_dropped, and output file with other variants is written to ./step3.avinput.hg38_
dbnsfp33a_filtered
NOTICE: Processing next batch with 42225 unique variants in 42225 input lines
NOTICE: Database index loaded. Total number of bins is 552168 and the number of bins to be scanned is 10956
```

图 11:进行功能注释，由于程序运行时间较长，后续将任务转移至计算节点完成

```
-rwx------ 1 teach80_pkuhpc teach80_pkuhpc      332 Mar 25 16:17 job.srp161737
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc        0 Mar 25 16:17 tab161737_1477.out
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc  1376939 Mar 25 18:08 step3.avinput.hg38_dbnsfp33a_filtered
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc  2224160 Mar 25 18:08 step3.avinput.hg38_dbnsfp33a_dropped
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc     1018 Mar 25 18:08 step3.avinput.log
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc     3364 Mar 25 18:08 tab161737_1477.err
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc  7942041 Mar 25 18:09 step3.avinput.hg38_multianno.txt
```

图 12:功能注释脚本输出的文件

step3.avinput.hg38_multianno.txt 的部分内容如图 13 所示。

```
Chr     Start   End     Ref     Alt     SIFT_score      SIFT_converted_rankscore        SIFT_pred       Polyphen2_HDIV_score    Polyphen2_HDIV_rankscore        Polyphen2_HDIV_pred     Polyphen2_HVAR_score    Polyph
en2_HVAR_rankscore      Polyphen2_HVAR_pred     LRT_score       LRT_converted_rankscore LRT_pred        MutationTaster_score    MutationTaster_converted_rankscore      MutationTaster_pred     MutationAssessor_sco
re  MutationAssessor_score_rankscore    MutationAssessor_pred   FATHMM_score    FATHMM_converted_rankscore      FATHMM_pred     PROVEAN_score   PROVEAN_converted_rankscore     PROVEAN_pred    VEST3_score     VE
ST3_rankscore MetaSVM_score   MetaSVM_rankscore       MetaSVM_pred    MetaLR_score    MetaLR_rankscore        MetaLR_pred     M-CAP_score     M-CAP_rankscore M-CAP_pred      CADD_raw        CADD_raw_rankscore
CADD_phred      DANN_score      DANN_rankscore  fathmm-MKL_coding_score fathmm-MKL_coding_rankscore     fathmm-MKL_coding_pred  Eigen_coding_or_noncoding       Eigen-raw       Eigen-PC-raw    GenoCanyon_score
  GenoCanyon_score_rankscore    integrated_fitCons_score        integrated_fitCons_score_rankscore      integrated_confidence_value     GERP++_RS       GERP++_RS_rankscore     phyloP100way_vertebrate phyloP100way
_vertebrate_rankscore   phyloP20way_mammalian   phyloP20way_mammalian_rankscore phastCons100way_vertebrate      phastCons100way_vertebrate_rankscore    phastCons20way_mammalian        phastCons20way_mammalian_r
ankscore      SiPhy_29way_logOdds     SiPhy_29way_logOdds_rankscore   Interpro_domain GTEx_V6_gene    GTEx_V6_tissue
chr1    19226   19226   T       A       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
  .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
    .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
chr1    19249   19249   G       A       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
  .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
    .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
chr1    19254   19254   G       C       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
  .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
    .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
chr1    19279   19279   C       T       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
  .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
    .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
chr1    103984  103984  G       A       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
  .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
    .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
chr1    104108  104108  A       G       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
  .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
    .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
chr1    264562  264562  C       T       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
  .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
    .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
chr1    601581  601581  T       C       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
  .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
    .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
chr1    866356  866356  G       T       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
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chr1    943911  943911  -       C       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
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chr1    1045883 1045883 G       A       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
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chr1    1091327 1091327 C       A       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
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chr1    1118072 1118072 G       A       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
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chr1    1309909 1309909 C       G       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
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chr1    1331929 1331929 C       T       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
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chr1    1354462 1354462 C       T       0.005   0.632   D       0.999   0.899   D       0.883   0.697   P       0.000   0.486   U       1       0.810   D       2.495   0.729   M       2.18    0.201   T       -3.03
  0.627   D       0.388   0.468   -0.854  0.517   T       0.140   0.460   T       0.075   0.722   D       6.278   0.867   29.0    0.999   0.976   0.869   0.460   D       c       0.496   0.406   1.000   0.747   0.76
7   0.994   0       4.43    0.528   5.845   0.689   0.777   0.313   1.000   0.715   0.196   0.216   16.059  0.805   Immunoglobulin V-set domain|Immunoglobulin subtype|Immunoglobulin-like domain|Immunoglobulin-like
fold  .       .
chr1    1390334 1390334 C       -       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
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chr1    1402084 1402084 C       G       .       .       .       .       .       .       .       .       .       .       .       .       1       0.090   N       .       .       .
  .       .       .       .       .       .       .       .       .       .       .       .       -1.495  0.017   0.003   0.599   0.062   0.955   0.649   D       c       0.429   0.173   1.000   0.747   0.38
3   0.051   0       3.31    0.369   1.689   0.371   0.020   0.144   1.000   0.715   0.948   0.423   6.436   0.209   .       .       .
chr1    1459308 1459308 G       C       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .       .
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图 13:功能注释结果

[讨论]

本次实验整体较为顺利，从测序后处理得到的.vcf 文件出发，完成了基因注释和功能注释，将测序发现的基因组中的突变与特定基因及其功能联系起来。进一步分析得到了基因变异频率的分布以及变异次数最多的基因，后续可进一步分析这些基因的变异对个体表型的影响。

[致谢]

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